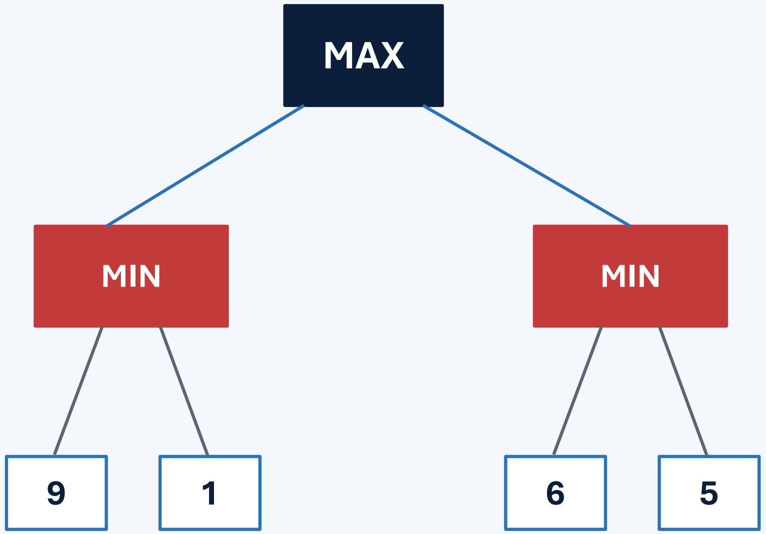
CompSci 471

Artificial Intelligence

Lesson 8: Adversarial and Stochastic Decision-Making

Good decisions depend not only on possible outcomes, but also on who—or what—controls them.



MIN: strategic response

CHANCE: probabilistic outcome

Fall 2026

Retrieval warm-up

Lessons 4–7 focused on representing and searching a single-agent problem.

In prior search problems, the agent chose actions and the environment followed known rules.

Question 1
What changes when another intelligent actor responds?

Question 2
What changes when the next outcome is random?

Question 3
Is “worst possible” the same as “most likely”?

Think-pair-share: who or what controls what happens after the agent acts?

Today's mission

By the end of Lesson 8, students should be able to...

1. **Explain minimax decision-making**
2. **Describe expectimax and stochastic environments**
3. **Compare adversarial vs probabilistic reasoning**

Performance target: evaluate a small decision tree from the leaves upward and justify whether MIN or CHANCE fits the situation.

Running scenario: aerial resupply

One aircraft chooses a delivery corridor; the environment determines the outcome.

Mission setup
Aircraft chooses one of two corridors
Higher utility = better outcome
Same possible outcomes, different model

Ridge Route
Fast and exposed
Can succeed dramatically
Can also fail badly

Valley Route
Slower and protected
Usually less extreme
May offer safer outcome

Before solving the tree, what must we know about the second decision point?

Opponent response or chance outcome?

Three kinds of decision environments

Algorithm selection begins with the environment model.

Deterministic single-agent
Known action result

Appropriate method:
Search / A*

Adversarial
Another decision-maker responds

Appropriate method:
Minimax

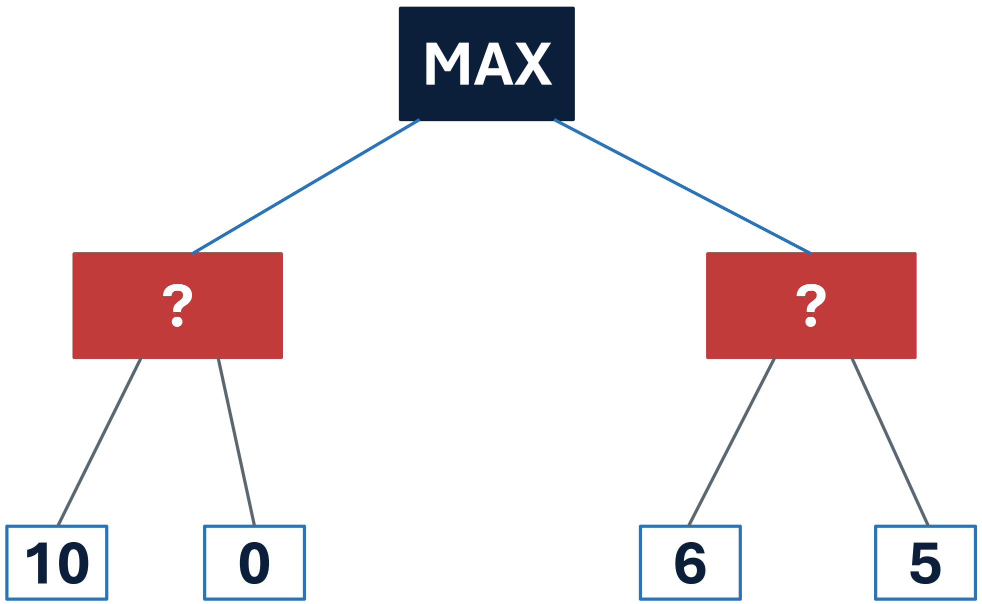
Stochastic
Outcome sampled from probabilities

Appropriate method:
Expectimax

Core question: who or what determines the next state?

Anatomy of a decision tree

Decision trees make the control structure visible.



MAX node
Our agent chooses the highest-valued action

MIN node
An opponent chooses what is worst for MAX

CHANCE node
Outcomes are combined using probabilities

Terminal utility
Numerical value of a completed outcome

Utility is always from one perspective

Unless stated otherwise, every value is measured from MAX’s perspective.

High utility

- Good for MAX
- mission success
- low loss
- strong payoff

Low utility

- Bad for MAX
- mission failure
- higher loss
- weak payoff

MIN’s role

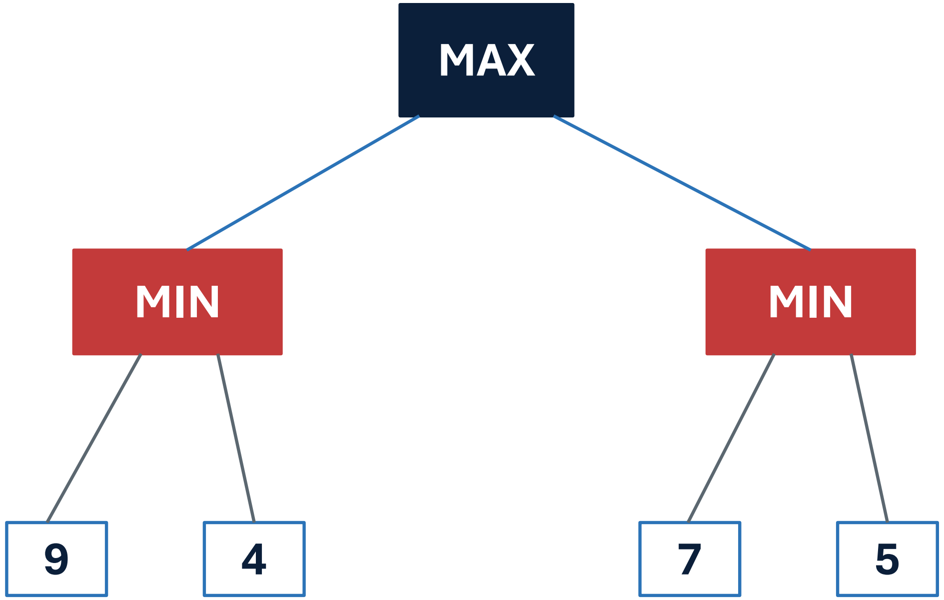
- Does not change the scale.
- Chooses the child with the lowest utility for MAX.

Keep one perspective throughout the whole tree.

Evaluate from the leaves upward

Back up values from terminal utilities to the root decision.

1. **Begin with terminal utilities**
2. **Evaluate the lowest internal nodes**
3. **Replace each subtree with its backed-up value**
4. **Continue toward the root**
5. **Choose the best root action**



Minimax models an intelligent opponent

MAX chooses well; MIN responds to make the outcome worse for MAX.

At MAX

Choose the maximum child value

V(n) = max(children)

At MIN

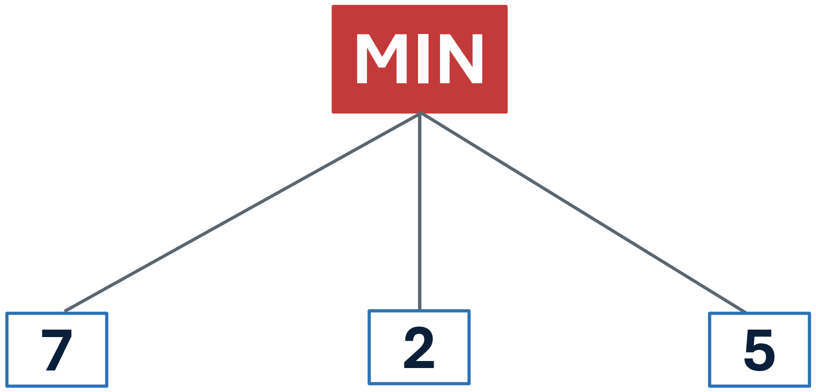
Choose the minimum child value

V(n) = min(children)

Minimax asks: what payoff can MAX guarantee against an optimal opponent?

MIN nodes choose the worst result for MAX

If an opponent can force a lower utility, MAX should plan for it.



$$\min(7, 2, 5) = 2$$

The opponent selects the child that is least favorable to MAX.

MAX nodes choose the best guaranteed branch

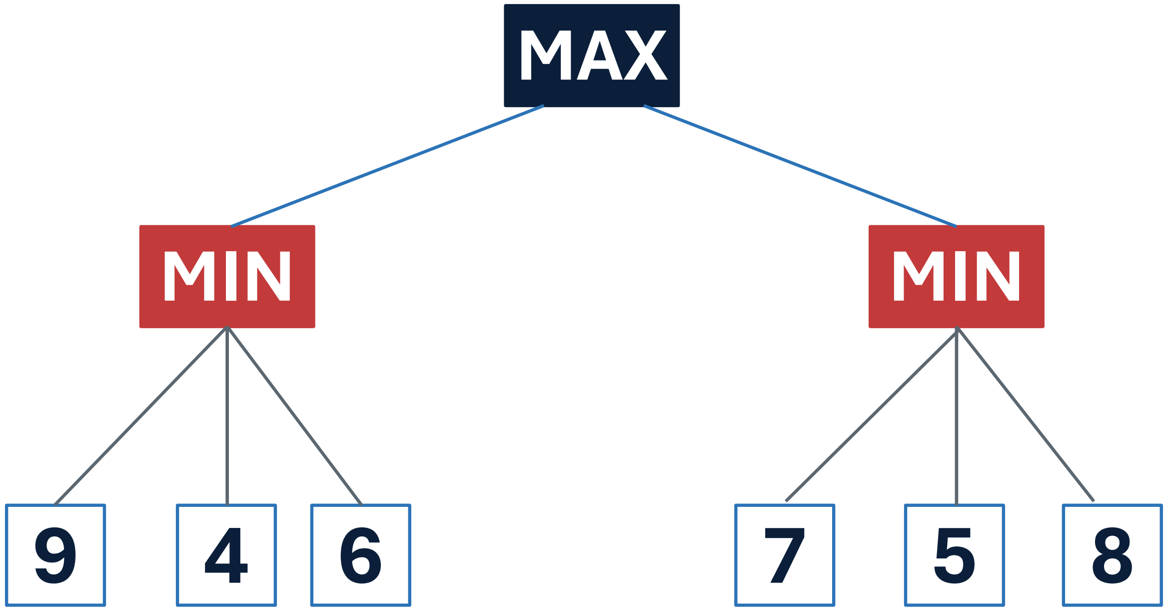
After MIN nodes are evaluated, MAX chooses the higher backed-up value.

max(2, 5) = 5

MAX chooses the action whose worst-case result is best.

Guided minimax trace

Evaluate each internal node, then choose at the root.



Calculate

Left MIN = min(9,4,6) = ?

Right MIN = min(7,5,8) = ?

Root MAX = max(left,right) = ?

What does the minimax root value mean?

The root value is the payoff MAX can guarantee against an optimal opponent.

It is...
Best guaranteed payoff under the modeled tree

It is not...
The average leaf value
The highest leaf anywhere
The most likely result

It assumes...
MIN is modeled as rational, adversarial, and able to choose the response

Minimax prepares for the best of the worst-case outcomes.

Minimax assumptions

Minimax is useful when poor outcomes are selected strategically.

Appropriate when

Another actor has conflicting objectives
The actor can respond to MAX
Utilities represent MAX preferences

Limitation

Minimax can be too conservative when bad outcomes are possible but not deliberately selected.

If the environment is not hostile, MIN may be the wrong model.

Expectimax models chance, not hostility

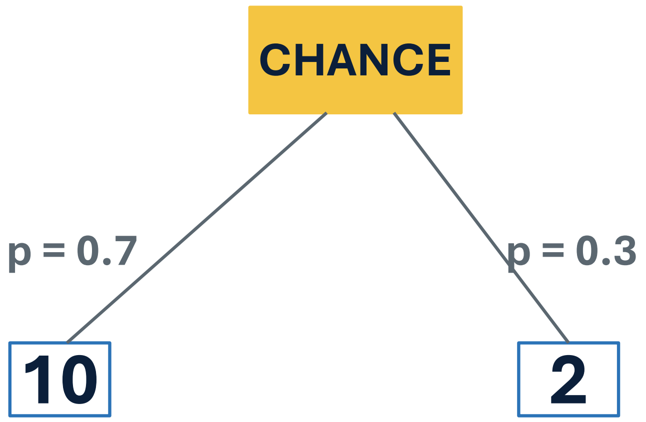
CHANCE nodes represent probabilistic outcomes rather than an opponent.

At MAX	At CHANCE
Choose the maximum child value	Compute probability-weighted value
$V(n) = \max(\text{children})$	$V(n) = \Sigma\, p_i\, V(i)$

Expectimax asks: which action has the best expected utility?

CHANCE nodes compute expected utility

The value is probability-weighted, not a guarantee.



$$0.7(10) + 0.3(2) = 7.6$$

Equal probabilities are a special case

Use an average only when outcomes are equally likely.

Two equal outcomes

$EV = (u_1 + u_2) / 2$

Do not assume

Probabilities must be provided or modeled. They should sum to 1.

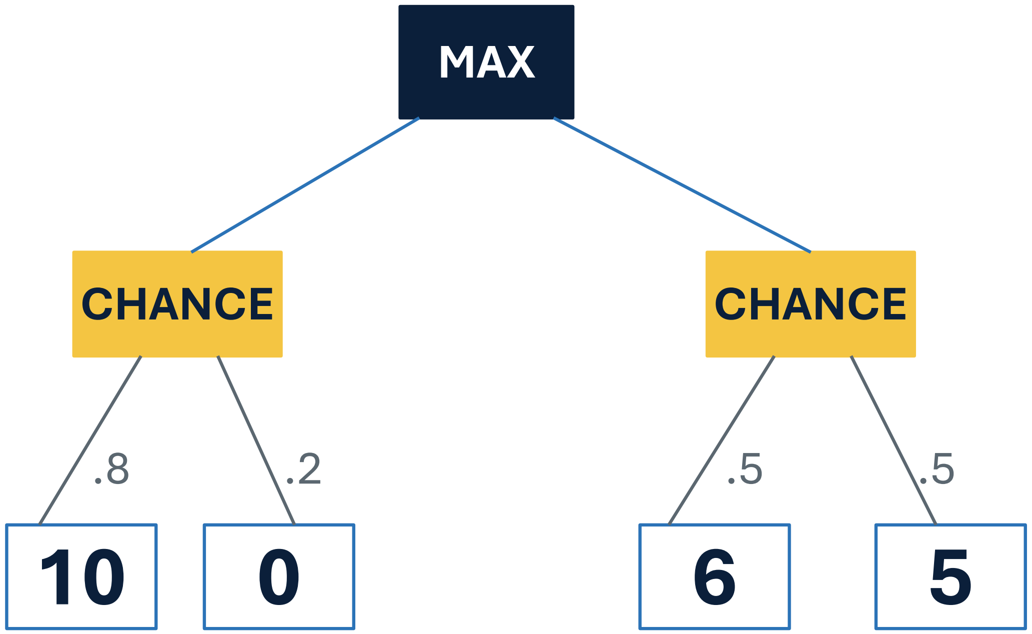
Interpretation

Expected value summarizes the probability-weighted utility.

Average-case reasoning is not worst-case reasoning.

Guided expectimax trace

Evaluate CHANCE nodes with probabilities, then choose at MAX.



Calculate

Left EV = .8(10) + .2(0) = ?

Right EV = .5(6) + .5(5) = ?

Root MAX = max(left,right) = ?

What does an expectimax root value mean?

The root value is the highest expected utility under the modeled probabilities.

It is...
Best probability-weighted value under the model

It is not...
A worst-case guarantee
The best possible result
A promise the value will occur

It depends on...
The correctness of the outcome probabilities

Expectimax chooses what performs best on average under the probability model.

Same leaves, different assumption

The tree structure can look the same while the rational decision changes.

Action A outcomes
10 and 0

Action B outcomes
6 and 5

Minimax
A → min(10,0) = 0
B → min(6,5) = 5
Choose B

Expectimax
A → .8(10)+.2(0) = 8
B → .5(6)+.5(5) = 5.5
Choose A

The environmental assumption changed—and so did the rational decision.

Minimax versus expectimax

The second-level node tells us what kind of reasoning is being used.

Feature	Minimax	Expectimax
Second-level node	MIN	CHANCE
Environment	Strategic opponent	Stochastic process
Backup operation	Minimum	Expected value
Root meaning	Best guaranteed payoff	Best expected payoff
Main risk	Too conservative	Bad probabilities

Choose the model

Name who or what controls the next outcome.

Opposing commander selects a countermeasure	Minimax
Weather produces outcomes with known probabilities	Expectimax
Robot follows a fixed map with known transitions	Search / A*
Cyber attacker adapts to the defender’s action	Minimax
Equipment failure follows estimated rates	Expectimax

Justification matters: strategic opponent, random outcome, or fixed transition?

Common mistakes

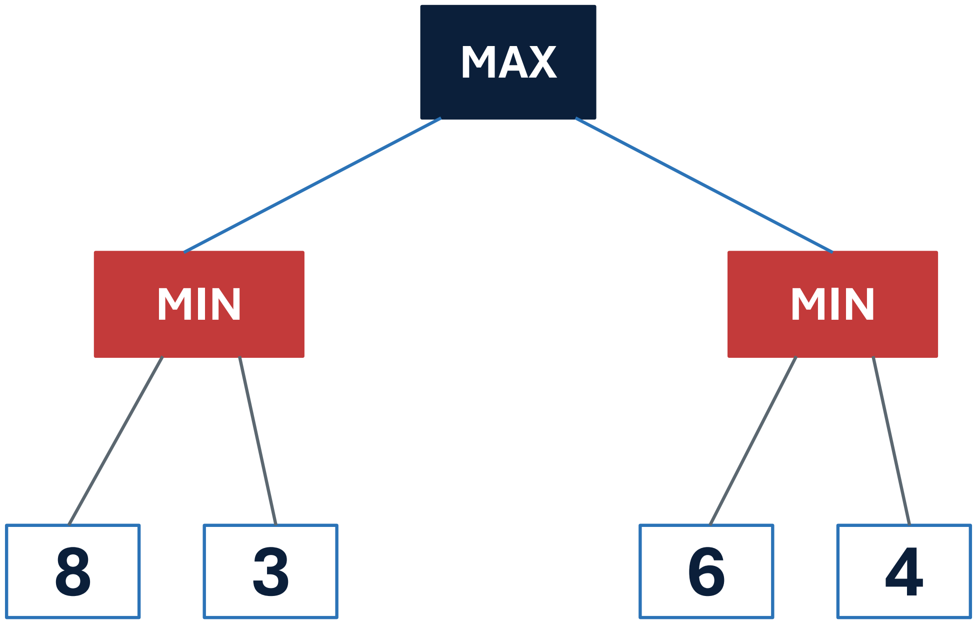
Correct the reasoning error, not just the final answer.

Mistake	Correction
Mistake 1 MAX chooses the largest leaf anywhere.	**Correction** Evaluate internal nodes first.
Mistake 2 MIN represents bad luck.	**Correction** MIN represents an adversarial actor.
Mistake 3 CHANCE chooses the worst outcome.	**Correction** CHANCE computes expected value.
Mistake 4 Expected value is guaranteed to occur.	**Correction** Expected value summarizes probabilities.
Mistake 5 Utilities switch perspective at MIN nodes.	**Correction** All values remain from MAX’s perspective.

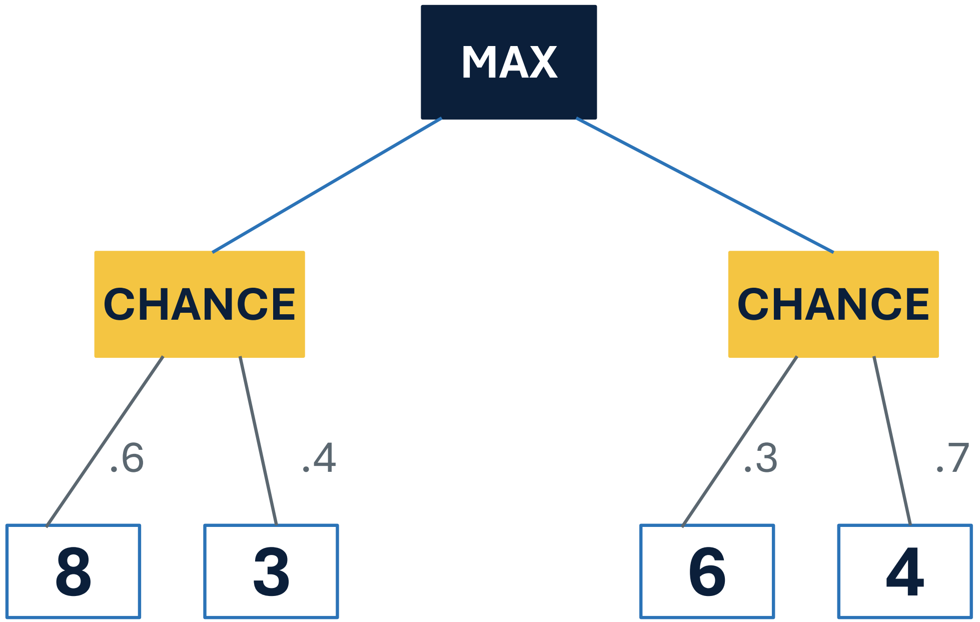
Pair decision-tree sprint

Evaluate two compact trees and explain the model assumption.

Tree A: minimax



Tree B: expectimax



Tasks: evaluate from leaves upward, identify MAX’s action, and state what assumption makes each model appropriate.

Synthesis

Decision-tree reasoning depends on the node type.

MAX node
choose the __________ value

MIN node
choose the __________ value

CHANCE node
compute the __________ value

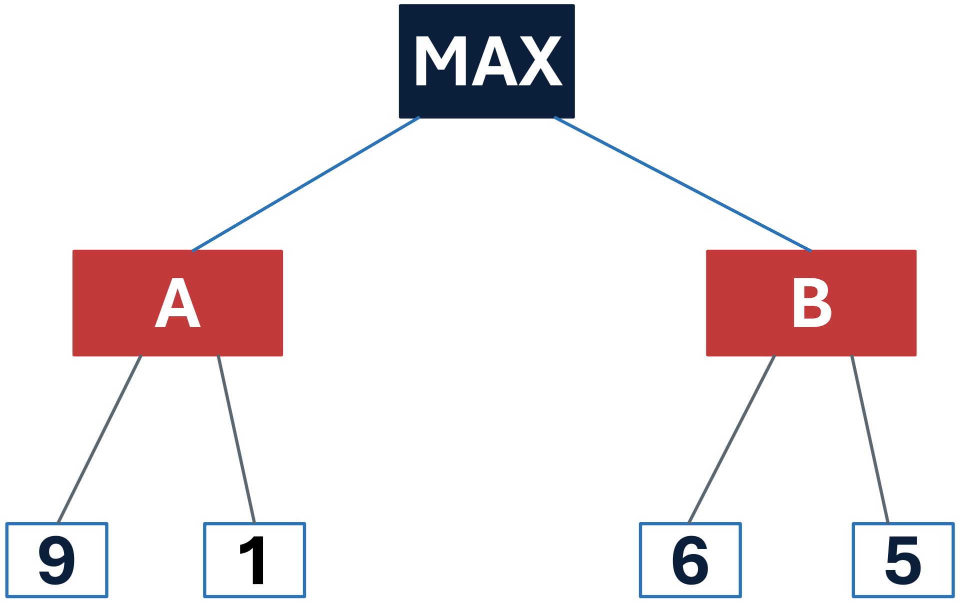
What single question determines whether a second-level node should be MIN or CHANCE?

Is the outcome selected strategically by an opponent, or generated probabilistically?

Exit Ticket

Apply minimax and expectimax to the same possible outcomes.

Outcomes
Action A: 9 and 1
Action B: 6 and 5



Answer:

1. Under minimax, which action does MAX choose?
2. If both outcomes are equally likely, which action does expectimax choose?
3. What does each root value represent?
4. Give one scenario appropriate for minimax.
5. Give one scenario appropriate for expectimax.

Reminder: HW2 due. Quiz 2 next class.